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Report

Ifood

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Introduction

In the digital marketplace, especially in food delivery and online retail, companies depend on efficient marketing to maintain engagement and drive growth. As competition grows and acquisition costs increase, personalized campaigns become vital. These campaigns help promote products, retain customers, and increase revenue. As a result, organizations adopt data-driven approaches to understand customer behavior and allocate marketing budgets well.

Blattberg, Kim & Neslin (2008) argue that broad and untargeted campaigns often yield weaker customer responses and can even be unprofitable. In contrast, precise customer targeting improves the return on investment of the marketing by enabling organizations to focus on customers with a higher likelihood of responding.

For companies with a large customer bases, such as those in the food delivery sector, understanding purchasing patterns and identifying customers with a strong chance of responding are especially important. Poorly targeted campaigns can result in unnecessary costs and even negative profits, whereas well-targeted campaigns can significantly improve business performance.

Furthermore, the growing availability of customer data creates an opportunity to apply advanced analytics and machine learning techniques to predict campaign responsiveness, personalize communication, and improve marketing efficiency.

Objectives

This study aims to develop a binary classification model to predict which customers are most likely to purchase a new product in an upcoming marketing campaign, with an emphasis on maximizing expected profit rather than solely optimizing predictive accuracy. In addition, the analysis seeks to understand key behavioral and demographic characteristics of responding customers, providing a customer profile to support the interpretation of the model.

Data

The dataset is derived from a real pilot marketing campaign conducted by the food company Ifood in which 2240 randomly selected customers were contacted by phone and offered a new product, published as an open challenge to those wanting to apply for a data analyst role within the company. The purchase decisions of the customers were recorded over the following months.

The data include socio-demographic and firmographic attributes describing each contacted customer. Table 1 summarizes the original set of features.

Using the variable `Year_birth`, we computed each customer's age. The `Dt_Customer` column was also used to calculate the number of days since the first interaction with the company, with the reference date being set to 100 days after the last

Feature	Description
Response	1 if customer accepted the offer in the last campaign, 0 otherwise
AcceptedCmp1	1 if customer accepted the offer in the 1st campaign, 0 otherwise
AcceptedCmp2	1 if customer accepted the offer in the 2nd campaign, 0 otherwise
AcceptedCmp3	1 if customer accepted the offer in the 3rd campaign, 0 otherwise
AcceptedCmp4	1 if customer accepted the offer in the 4th campaign, 0 otherwise
AcceptedCmp5	1 if customer accepted the offer in the 5th campaign, 0 otherwise
Complain	1 if customer complained in the last 2 years
Year_Birth	customer's year of birth
Dt_Customer	date of customer's enrollment with the company
Marital_Status	customer's marital status
Education	customer's level of education
Income	customer's yearly household income
Kidhome	number of small children in customer's household
Teenhome	number of teenagers in customer's household
MntFishProducts	amount spent on fish products in the last 2 years
MntFruits	amount spent on fruits in the last 2 years
MntGoldProducts	amount spent on *gold* products in the last 2 years
MntMeatProducts	amount spent on meat products in the last 2 years
MntSweetProducts	amount spent on sweet products in the last 2 years
MntWines	amount spent on wines in the last 2 years
NumCatalogPurchases	number of purchases made using catalogue
NumDealsPurchases	number of purchases made with discount
NumStorePurchases	number of purchases made directly in stores
NumWebPurchases	number of purchases made through company's website
NumWebVisitsMonth	number of visits to company's website in the last month
Recency	number of days since the last purchase
Z_CostContact	amount spent by the company to contact a customer
Z_Revenue	revenue generated if the customer accepts the offer

Table 1: Initial variables available in the data

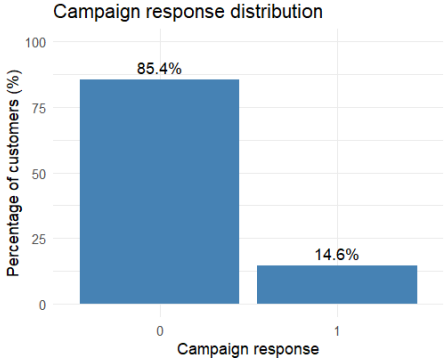
customer join date within the dataset, so as to line up with the values in the `Recency` variable, this was then stored in the new feature `Customer_Days`. In addition, aggregate measures were also created. The feature `MntTotal` represents the total spending of each customer, obtained by summing the four monetary expenditure features. Similarly, the `TotalCmp` variable represents the total number of accepted campaigns and was obtained by summing all campaign acceptance indicators for each customer.

Finally, the columns `ID`, `Year_birth`, `Dt_Customer`, `ZCostContact` and `ZRevenue` were dropped since they didn't add any valuable information (`ZCostContact` and `ZRevenue` are 3 and 11 for every customer, respectively). An initial inspection of the dataset revealed the presence of duplicate records that required cleaning as well as missing and non compliant responses as would be expected from real word data obtained from customer input. Among these answers are multiple customers who state their age to be > 110 years or provide non-informative answers to the marital status ("YOLO","Absurd"), the former was solved by taking the mean age for customers with the same education and marital status, while the latter was solved by assigning these values to the more broader "Single" factor. Some missing values were also detected within the Income feature and were resolved by substituting with mean income for customers with the same education, same marital status, and ± 10 years of age.

Exploratory Data Analysis

The target indicates whether a customer accepted the offer during the pilot marketing campaign. As shown in Figure 1a, the observed response rate is approximately 14.6%, resulting in a strongly imbalanced classification problem. This imbalance has economic implications. Based on the pilot campaign, contacting a customer has a cost of 3MU, while a successful sale generates a profit of 8MU. Therefore, under indiscriminate targeting, the expected profit per contacted customer is approximately -1.84MU. This implies that contacting all customers leads to a negative expected profit, confirming that mass targeting is economically inefficient.

Taking this into account, we have examined the relationship between customer characteristics and campaign response in order to identify variables that may support profitable targeting decisions. Among categorical variables, past campaign behavior emerges as one of the most informative factors, as shown in Table 1b. Customers who accepted previous campaigns exhibit substantially higher response rates in the current campaign, with acceptance increasing sharply with the number of positive responses in the past. Although customers with multiple past acceptances represent a small fraction of the sample, their response rates are markedly higher than the baseline, making this variable economically relevant from a profit maximization perspective.



(a) Distribution of Campaign Response

Variable	Level	Response rate	Share(%)
TotalCmp	0	0.0786	79.4
	1	0.305	14.5
	2	0.507	3.71
	3	0.8	1.98
	4	0.9	0.495

(b) Response rates by number of previously accepted campaigns

Figure 1

The composition of the household also shows meaningful associations with response. In particular, customers without teenagers or children at home are more likely to accept the offer. Socio-demographic variables such as education display weaker but consistent gradients, with higher education levels associated with higher response rates, while other categorical variables such as marital status and complaint history exhibit less stable or interpretable patterns.

Variable	Level	Response rate	Share(%)
Teenhome	0	0.206	51.9
	1	0.0792	45.6
	2	0.1	2.48

(a) Response rates by number of teens in chustomer's household

Variable	Level	Response rate	Share(%)
Kidhome	0	0.172	57.7
	1	0.113	40.1
	2	0.0465	2.13

(b) Response rates by number of kids in customer's household

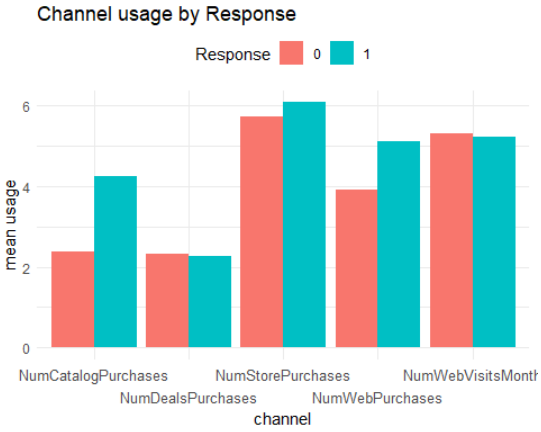
Figure 2

Numerical predictors also display some individual associations with campaign response, as shown in Table 3b. Responding customers tend to have higher historical spending and more recent purchase activity, as reflected by higher values of total monetary spending and lower values of recency. In contrast, age shows no meaningful difference between responders and non responders. Income and customer tenure exhibit moderate shifts that are likely correlated with spending and recency, indicating potential redundancy.

In addition, responders differ in their purchasing activity across sales channels. As illustrated in Figure 3a, customers who accepted the campaign exhibit higher average numbers of catalog, store and web purchases, while differences in web visit frequency are relatively limited. This suggests that campaign response is associated with active transactional behavior rather than mere exposure or browsing intensity.

Overall, the distributions of responders and non responders largely overlap across individual predictors, indicating that no single variable provides strong discrimination on its own. Instead, predictive power is expected to arise from the joint contribution of multiple weak signals.

To assess potential redundancy among predictors, pairwise correlations were examined for numerical variables and selected binary indicators. The resulting correlation matrix, summarized in Figure 4, reveals several well defined groups of highly related variables, reflecting overlapping dimensions of customer behavior and lifecycle.



(a) Campaign response by channel usage

Variable	Responders	Non Responders
Income (MU)	64506	50344
MntTotal (MU)	1000	273
Age (Years)	44	44
Recency (days)	30	52
Customer_days (days)	586	426

(b) Median of Numerical Variables by Response

Figure 3

Some well defined blocks of highly correlated variables emerge, reflecting overlapping dimensions of customer behavior and lifecycle.

A first block consists of spending related variables, including expenditures across individual product categories and their aggregate measure. These variables exhibit strong correlations, partly reflecting consistent purchasing behavior across products, but also partly arising from feature construction since total monetary spending is defined as the sum of category level expenditures. Beyond this feature construction component, spending variables also display moderate to strong positive correlations with purchase counts across sale channels such as web, catalog and store purchases. In contrast, web browsing is negatively correlated with the expenditure variables. These associations reflect underlying behavioral patterns in the data, in which customers with higher transactional activity tend to spend more overall. Similarly, income is positively correlated with spending measures and with the usage of most channels.

Behavioral variables capturing past campaign acceptance form a distinct correlated block. Individual campaign response indicators are associated with each other, their aggregate measure and the current campaign response.

Finally, household composition variables show correlation with spending and behavioral features. The number of children and teenagers is negatively correlated with consumption related variables and income, reflecting overlapping family related constraints that may contribute to multicollinearity in linear models.

Overall, the presence of correlated predictors indicates that multiple variables convey overlapping information rather than independent signals.

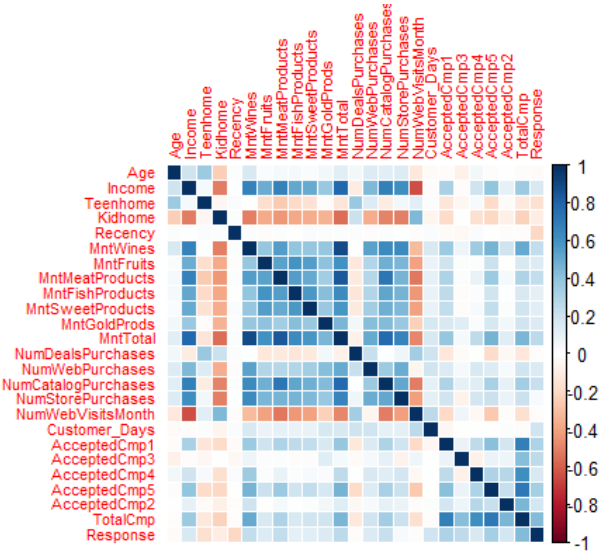


Figure 4: Correlation heatmap

Exploratory PCA: structure and interpretation

To address the possibility that several predictors carry overlapping information (particularly the "spending" block identified in the correlation analysis), we applied PCA with dimensionality reduction in mind.

Only numerical variables were considered, because, although categorical variables can be included in PCA by means of dummy coding, this practice is generally discouraged in classical PCA, as dummy variables introduce artificial distances and may distort the variance structure, reducing interpretability. In this analysis, PCA is therefore applied exclusively to numeric predictors, where multicollinearity is most pronounced.

Because the numeric predictors are measured on very different scales (e.g., Income in tens of thousands, monetary spending variables, purchase counts, and Recency in days), PCA was performed on standardized variables (centered and scaled), ensuring that each variable contributes equally to the computation of the principal components and prevents variables with larger numerical ranges from dominating the results.

Explained variance and component retention

To determine the number of components to retain, several complementary criteria were considered.

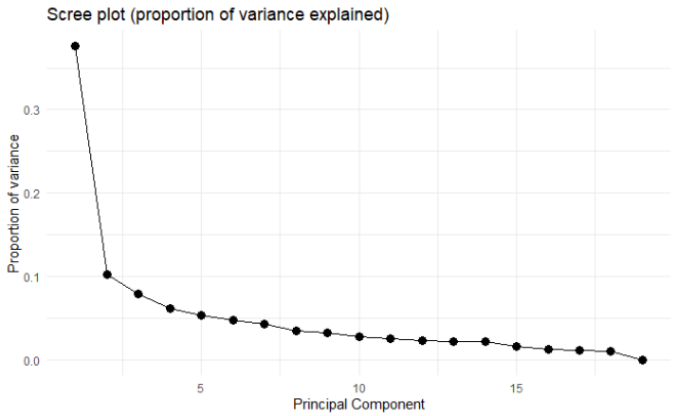


Figure 5: PCA Scree Plot

The scree plot exhibits a clear elbow after the first principal component, followed by a gradual decay. This suggests that PC1 captures a dominant latent factor, while subsequent components add more specific and nuanced information or possibly just noise.

```
=== PCA retention criteria (TRAIN PCA) ===
k (>=80% variance): 9   | CumVar = 83.03 %
k (>=90% variance): 12  | CumVar = 90.57 %
k (Kaiser eig>=1): 5    | CumVar = 67.29 %
```

Figure 6: PCA retention criteria

The cumulative explained variance analysis shows that:

- approximately 83.03% of the total variance is explained by the first 9 components;
- approximately 90.57% is explained by the first 12 components.

Since the PCA was performed on standardized variables, the Kaiser criterion was also applied. According to this criterion, 5 principal components should be retained (67.29% of variance explained). For interpretative purposes, the Kaiser criterion was adopted, as it yields a small number of components, which retain a substantial amount of variability and remain meaningful and easy to interpret.

Interpretation of the principal components

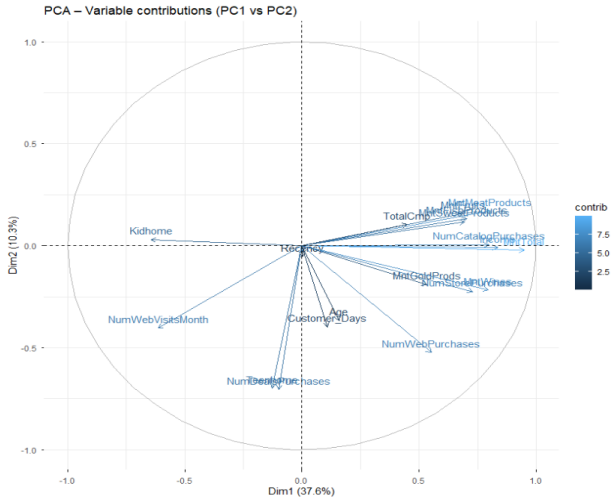
The interpretation of the principal components was based on the inspection of the loading matrix, focusing on variables with the largest absolute loadings for each component:

- PC1 (37.61% of variance) aggregates how much a customer spends;
- PC2 (10.26% of variance) is not about how much people spend, but how and why they engage, separating deal-seeking online users from more stable purchasing profiles;

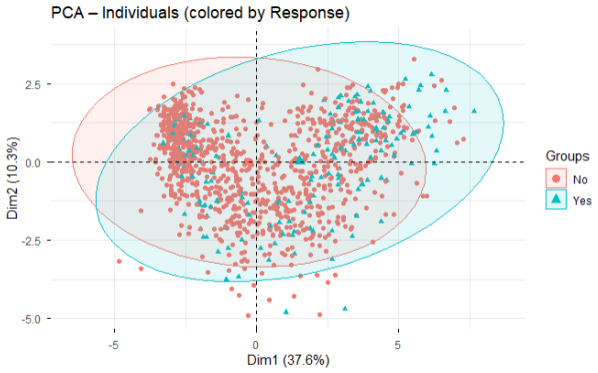
- PC3 (7.91% of variance) distinguishes customers based on demographic and household context, not directly on purchasing volume;
- PC4 (6.14% of variance) is a pure marketing responsiveness axis;
- PC5 (5.37% of variance) isolates time since last purchase, making it conceptually distinct and easy to interpret.

Beyond the first few components, interpretability decreases as later components tend to capture increasingly specific and less stable patterns, supporting the decision to focus on the first components for both interpretation and prediction.

Visualization in the reduced space



(a) PCA loadings plot of contribution and correlation of variables with PC1 and PC2



(b) PCA score plot of individuals on PC1 and PC2

Figure 7: PCA results: individual scores and variable contributions on the first two principal components.

The variables factor map for PC1 and PC2 highlights clear clusters of correlated variables, particularly among spending and purchasing indicators, supporting the interpretation of PC1 as an overall spending dimension. The individuals factor map shows partial separation between responders and non-responders along PC1. While there is considerable overlap, customers with higher PC1 scores tend to have a higher likelihood of acceptance, suggesting that purchasing intensity is positively associated with campaign response. This indicates that while PCA captures relevant structure, class separation is not fully achieved in low-dimensional projections.

PCA-Based Logistic Regression

Model Training

To address multicollinearity among the original predictors and reduce dimensionality, a logistic regression model was trained using principal component scores instead of the original variables. The PCA was estimated exclusively on the training set, and the same PCA transformation was subsequently applied to the validation and test sets, ensuring that no information from the hold-out samples leaked into the model training process.

The first k PCs were retained and used as predictors in the logistic regression model, with the response variable indicating whether the customer accepted the campaign. Model fit was assessed using McFadden's pseudo-R^2.

Although the Kaiser criterion suggests retaining five components for interpretability, additional values of k were evaluated for predictive purposes. Increasing the number of components to k=9 (80% variance criteria) led to a marked improvement in McFadden's pseudo-R^2 (from approximately 0.24 to 0.31), indicating that components explaining smaller proportions of variance still carry relevant discriminative information. Increasing further to k=12 (90% variance criteria) resulted in only marginal gains (approximately 0.006). Based on this trade-off between model fit and complexity, k=9 was selected for the final predictive model.

The PCA-based logistic regression achieved a McFadden pseudo-R^2 of 0.312, indicating a strong improvement over the intercept-only model. In logistic regression contexts, such values are typically considered indicative of a well-fitting model, confirming that the retained components capture substantial discriminative information.

Profit-Based Cutoff Selection

Our models output the probability $p = \hat{P}(y_i = 1|\mathbf{x}_i)$, to align the predictions with our business objectives we can thus optimize the theoretical decision threshold such that it maximizes the profit of the campaign by contacting a customer every time the expected value of such is positive, knowing that contacting a non paying customer yields a profit of -3 while a paying one yields +8.

$$E[\text{contacting customer } \mathbf{x}_i|p] = 8p - 3(1-p) = 11p - 3$$

$$E[\text{contacting customer } \mathbf{x}_i|p] > 0 \Leftrightarrow 11p - 3 > 0 \Leftrightarrow p > \frac{3}{11} = 0.273$$

So we should contact our customers whenever our model's predicted probability output is greater than 0.273.

However, our models may not always be perfectly calibrated, causing the probability outputs to be skewed, thus an empirical cutoff was calculated on the validation set by evaluating the resulting profit across a grid of thresholds between 0 and 1, choosing the threshold that maximizes the validation set profit. In this case the resulting cutoff was 0.233, indicating a reasonable level of calibration when compared with the theoretical results.

Test Set Performance: Empirical and Theoretical Cutoff

The empirical profit-optimal cutoff was applied to the test set to evaluate out-of-sample performance. The resulting ROC curve yielded an AUC of 0.917, reflecting excellent discriminatory ability. At this cutoff, the model achieved a balanced accuracy of approximately 0.82, with a sensitivity of 72.6% and a specificity of 90.9%, indicating an effective balance between identifying responders and avoiding unnecessary targeting. The total profit obtained on the test set was 307 units, confirming that the cutoff selected on the validation set generalizes well to unseen data and leads to substantial economic gains.

For comparison, model performance was also evaluated using the theoretical cutoff. While overall accuracy remained high (89.1%), sensitivity decreased relative to the optimal cutoff (67.1%), resulting in fewer correctly identified responders. Under this fixed threshold, the total profit on the test set was 299 units, lower than the profit achieved using the optimized cutoff.

This result highlights how incorporating a profit-based cutoff substantially improves decision quality, underscoring the importance of aligning predictive modeling with business objectives rather than relying solely on traditional performance metrics.

Model selection

The primary objective of this project is to maximize the expected profit of a marketing campaign, rather than to optimize purely statistical performance measures such as accuracy. In this context, a profit-based evaluation is more appropriate, since the consequences of classification errors are asymmetric: correctly targeting a responding customer yields a gain of 8 units, while incorrectly targeting a non-responding customer incurs a cost of 3 units. Customers who are not targeted generate neither cost nor profit but may pose as lost opportunities.

Although several classification models were estimated and evaluated, an in-depth analysis is presented only for the PCA-based logistic regression model. Focusing on this model allowed a detailed discussion of component selection, interpretation, and the integration of PCA as a dimensionality-reduction step within a classification model. Nevertheless, all remaining models were subjected to the same evaluation pipeline: model training on 60%, profit-based cutoff selection using the validation set (15% of the dataset), and performance assessment on the test set (25% of the dataset) under both optimal and theoretical cutoffs, ensuring a fair and consistent comparison across approaches.

Overall, the results in Table 2 indicate that while several models demonstrate a strong discriminatory ability, as reflected by AUC values above 0.90, the choice of model based solely on predictive accuracy would be insufficient in this application. Instead, model evaluation must be aligned with the business objective of profit maximization.

The highest AUC is obtained by the random forest followed closely by the Lasso with interactions and standard Lasso. This indicates that nonlinear and regularized approaches slightly improve ranking ability, although these gains are modest. In addition, profit outcomes differ substantially even among models with similar AUC and balanced accuracy. For example, the Lasso with interactions improves AUC relative to the standard Lasso but results in lower profit, suggesting that additional model complexity does not translate into economic gains under the adopted payoff structure.

Finally, among all the models evaluated, the Lasso logistic regression achieved the highest total profit on the test set (320 units). Although other models achieved competitive predictive performance, none exceeded the Lasso model in terms of realized profit under the adopted payoff structure and cutoff selection procedure.

Model	AUC	Cutoff	Sensitivity	Specificity	PPV	Balanced Accuracy	Profit
Logistic	0.9111	0.14	0.7945	0.8329	0.4462	0.8137	248
		3/11	0.6849	0.9327	0.6329	0.8088	313
Lasso	0.9198	**0.284**	0.6575	0.9490	0.6857	0.8032	318
		3/11	0.6712	0.9443	0.6712	0.8078	**320**
Lasso with interactions	0.9241	0.227	0.6712	0.9281	0.6125	0.7996	299
		3/11	0.5616	0.9466	0.6406	0.7541	259
PCA-based logistic	0.9172	0.233	0.7260	0.9095	0.5761	**0.8178**	307
		3/11	0.6712	0.9281	0.6125	0.7997	299
LDA with equal priors	0.9114	0.75	0.5890	**0.9582**	**0.7049**	0.7736	290
		3/11	**0.8356**	0.7541	0.3653	0.7948	170
Random forest	**0.9258**	0.325	0.6986	0.9304	0.6296	0.8145	318
		3/11	0.7260	0.8910	0.5300	0.8085	283
MLP	0.8920	0.436	0.5068	0.9466	0.6167	0.7267	227
		3/11	0.6301	0.9258	0.5897	0.7779	272

Table 2: Model comparison

In addition to its superior profit performance, the Lasso model offers several methodological advantages. Lasso performs automatic variable selection, leading to a sparse and more interpretable model. This reduces overfitting risk and facilitates managerial interpretation of the drivers behind campaign response. For these reasons, the Lasso logistic regression was selected as the final model for subsequent analysis and explanation.

An examination of the estimated Lasso coefficients provides further insight into the determinants of campaigns response. The most influential predictors are related to previous campaign engagement. The number of past accepted campaigns exhibits the largest positive coefficient, indicating a strong persistence effect. Purchasing behavior, particularly channel usage, also plays a key role. A higher number of web purchases is associated with a higher probability of response. In contrast, a greater number of in store purchases is negatively related to response.

With respects to demographics, age has a small positive association with response probability. Education exhibits a clear gradient: customers with higher educational attainment are more likely to respond. Marital status effects suggest that single customers are more responsive than married or cohabiting individuals. Household composition further influences response. The presence of teenagers in the household reduces response likelihood, possibly reflecting tighter time or budget constraints.

Relationship duration has a small positive effect, indicating that longer tenured customers are more receptive to outreach, while higher recency values are associated with lower response probability, consistent with a reactivation pattern among less recently active customers.

Additionally, several variables, such as income, total monetary spending and multiple product category expenditures are excluded from the model through coefficient shrinkage. Their removal suggests that these factors do not provide additional predictive power once engagement history and purchasing behavior are accounted for. Taken together, the Lasso logistic regression not only delivers the highest profit among all evaluated models but also yields a coherent and interpretable structure that aligns closely with observed customer behavior.

Finally the empirical cutoff of 0.284 that was obtained for Lasso Regression is, out of all the models, the closest to the theoretical cutoff of 0.273. This indicates a very well calibrated model that can more predictably generalize to other data.

Another model that achieved similar profits results was the random forest. Although the two models rely on fundamentally different mechanisms, both appear well suited to the underlying structure of the data and the profit oriented decision rule.

The strong performance of the random forest can be attributed to its ability to capture nonlinear relationships among predictors.

The random forest model was tuned using a grid search over key hyperparameters that govern model complexity and tree diversity: the number of trees (num.trees), the number of candidate variables considered at each split (mtry), and the minimum terminal node size (min.node.size). A predefined grid of values was evaluated to explore a range of model configurations, balancing predictive flexibility and generalization.

For each combination of hyperparameters, the model was trained on the training set and evaluated on the validation set using predicted response probabilities. Rather than optimizing a standard accuracy-based metric, model performance was assessed using the same profit-based objective adopted throughout the analysis. Specifically, for each fitted model, profits were computed over a range of decision thresholds, and the maximum attainable profit on the validation set was recorded.

The hyperparameter configuration yielding the highest validation-set profit was selected for final model estimation and

evaluation on the test set. This tuning strategy ensures that the random forest model is calibrated to the underlying economic objective, rather than to purely statistical performance measures, and allows for a fair comparison with the Lasso logistic regression under a consistent payoff structure.

Importantly, the profit performance of the random forest is very close to that of the Lasso model, despite differences in model structure and interpretability. This suggests that both approaches successfully identify a similar subset of high-value customers, even though they do so via different functional forms. The comparable profit levels indicate that the underlying signal in the data is strong and can be exploited either through regularization and sparsity (Lasso) or through flexible, nonparametric learning (random forest).

For the MLP the hyperparameters (Layer size, weight decay, number of iterations) were tuned with a random grid search with cross validation, with the best results being obtained for 35 neurons in one layer. The overall results were sub-par with very poor calibration evident by the massive drop in profit from using the empirical cutoff calculcated on the validation set.

We thus conclude that the Lasso Logistic Regression was the best performing and calibrated model.

PAC and Farness Analysis

While global performance measures such as AUC and profit quantify how well a model performs on average, they provide limited insight into why individual predictions are correct or incorrect. To address this limitation, PAC and Farness were applied to the final selected model.

These measures are not intended for model comparison, but rather for post-hoc explanation of a chosen classifier, allowing a deeper understanding of prediction uncertainty and data structure at the individual observation level. Although PAC and Farness are often particularly informative for complex or non-linear classifiers, their application to a regularized linear model such as Lasso logistic regression remains appropriate. In this context, the analysis serves primarily as a diagnostic tool to confirm model stability, identify regions of intrinsic class overlap, and verify that classification uncertainty arises from data structure rather than model instability.

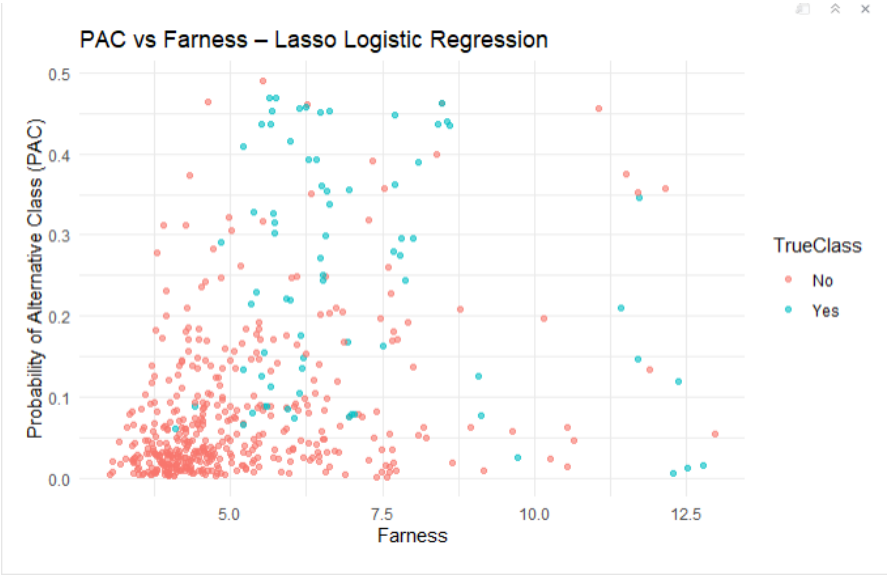


Figure 8: PAC vs Farness class map for LASSO

Interpretation of the PAC-Farness Class Map

The PAC–Farness results for the Lasso logistic regression model indicate that the majority of observations are classified with high confidence and are representative of their respective classes. Most customers exhibit very low PAC values, suggesting that the model assigns them to their predicted class with little ambiguity. This concentration of points at low PAC levels indicates strong separation for typical customer profiles.

A large proportion of observations also display relatively low farness values, implying that they lie close to the centroid of their true class in the standardized feature space. These cases correspond to well-represented and stable customer profiles for which the model's predictions are both confident and consistent. This pattern is particularly evident for the "No" class,

which dominates the dataset and forms a dense cluster of low-PAC, low-farness observations.

A smaller number of customers exhibit high farness while maintaining low PAC values. These observations correspond to atypical customer profiles that differ substantially from the average behaviour of their class, yet are still classified with high confidence. This suggests that the Lasso model generalizes beyond the core of each class and can handle unusual patterns without introducing classification uncertainty.

Some observations show moderate to high PAC values at moderate farness levels, reflecting genuine ambiguity in classification. These cases are likely driven by overlap between responder and non-responder profiles in certain regions of the feature space. Importantly, these ambiguous observations are not extreme outliers, indicating that uncertainty arises from structural similarity between classes rather than from noise or data quality issues.

Very few observations simultaneously exhibit high PAC and high farness. The scarcity of such cases suggests that extreme outliers or potentially mislabeled observations are rare in the data, further supporting the robustness of the selected model.

A notable pattern is that a non-negligible number of observations from the "Yes" class exhibit relatively high PAC values. This indicates higher classification uncertainty for responders, reflecting substantial overlap between responder and non-responder profiles. Such behaviour is typical in response modelling contexts and suggests that ambiguity arises from intrinsic class similarity rather than model instability.

Overall, the PAC–Farness analysis confirms that the selected Lasso logistic regression model behaves in a stable and interpretable manner. The model performs confidently for the majority of customers, handles atypical cases without excessive uncertainty, and concentrates ambiguity in regions of genuine class overlap. These findings complement the profit-based evaluation and reinforce the suitability of the Lasso model for deployment in this application.

Conclusion

The presented study addressed the problem of identifying customers with a high likelihood of responding to a marketing campaign, with the explicit objective of maximizing expected profit rather than optimizing purely statistical performance measures. Using data from a real pilot campaign conducted by Ifood, we performed a comprehensive analysis combining exploratory data analysis, feature engineering, dimensionality reduction, and supervised classification.

The exploratory analysis revealed a strongly imbalanced response variable and showed that indiscriminate targeting leads to massively negative expected profit. Behavioural variables, particularly past campaign acceptance and purchasing activity across channels, emerged as the most informative predictors, while socio-demographic variables played a secondary but interpretable role. Correlation analysis highlighted substantial redundancy among numeric predictors, motivating the use of dimensionality reduction techniques.

A paradigmatic example of this kind of methods is Principal Component Analysis, which proved effective in summarizing highly correlated numerical variables into a smaller number of interpretable components. A PCA-based logistic regression achieved strong discriminatory performance and confirmed that spending intensity, marketing responsiveness, and purchase recency are central dimensions underlying campaign response. However, while PCA improved model stability and interpretability, it did not outperform alternative methods in terms of economic outcomes.

A range of supervised learning models was evaluated using a profit-based decision framework. The results demonstrated that traditional metrics such as accuracy or AUC alone were insufficient for model selection in this context. Instead, aligning the classification threshold with the economic payoff structure was critical to achieving positive returns. Among all evaluated models, the Lasso logistic regression achieved the highest profit on the test set while offering the best calibration and a sparse and interpretable representation of the response drivers. Its coefficient structure highlighted the dominant role of prior campaign engagement and active purchasing behaviour, while excluding several demographic and spending variables that provided no additional predictive value once engagement history was accounted for.

Post-hoc analysis using PAC and Farness confirmed that the selected model is stable and confident for the majority of customers, with classification uncertainty largely confined to regions of genuine overlap between responders and non-responders rather than to outliers or data quality issues.

Despite these encouraging results, several limitations should be acknowledged. The analysis relies on observational data from a single pilot campaign, and the response variable captures observed acceptance rather than the causal effect of being contacted. Moreover, data cleaning and imputation choices may have influenced the results, and the analysis assumes homogeneous and fixed contact costs and revenues across customers. In addition, model validation was limited to a single dataset, leaving generalization to future campaigns uncertain.

Future work should therefore focus on casual uplift modelling and randomized experimentation to estimate incremental effects, improved probability calibration, and the incorporation of richer temporal and behavioural features. Validating the approach across multiple campaigns and possibly employing clustering to segregate the customer base so as to better understand each demographic's behaviour.